GHG Statement Report

Version 1.7

Date: August 19th, 2026

Information requirements for verification are as follows:

- GHG Statement Report (this document) containing information relating to the GHG Statement
- GHG Statement calculations (provided separately)
- Supporting information including copies of raw data used

An appropriate level of information should be provided with a clearly referenced and transparent audit trail of decision making, assumptions, explanations, such that a verifier can trace all inputs, outputs and decision making. All data sources and assumptions must be clearly referenced, transparent and traceable.

After this document has been completed for the first time, it may be acceptable to refer back to relevant sections for subsequent Reporting Periods, for example if there have been no changes to the GHG Assessment scope, data collection methodology, calculation methodology or assumptions.

There is no word limit for answers in sections below.

Section A – General Project Information

General Information	
Name of practitioner who prepared the GHG Statement, and relevant competencies. This should include an overview of relevant qualifications and experience undertaking GHG assessments	Kieran Lewko, NULIFE GreenTech Director of Carbon Markets
Name of Project / link to Project page on Isometric Registry	Project — Isometric
Date of report	August 19th, 2026
Please specify the Reporting Period this GHG Statement has been prepared for (the Reporting Period describes the time period over which the carbon removal activity assessed in the GHG Statement occurred)	GHG Statement: 26 Jun 2026 to 22 Jul 2026 ggs_1M0B8VWHZ1S0JJAK
Relevant project information if not included	N/A

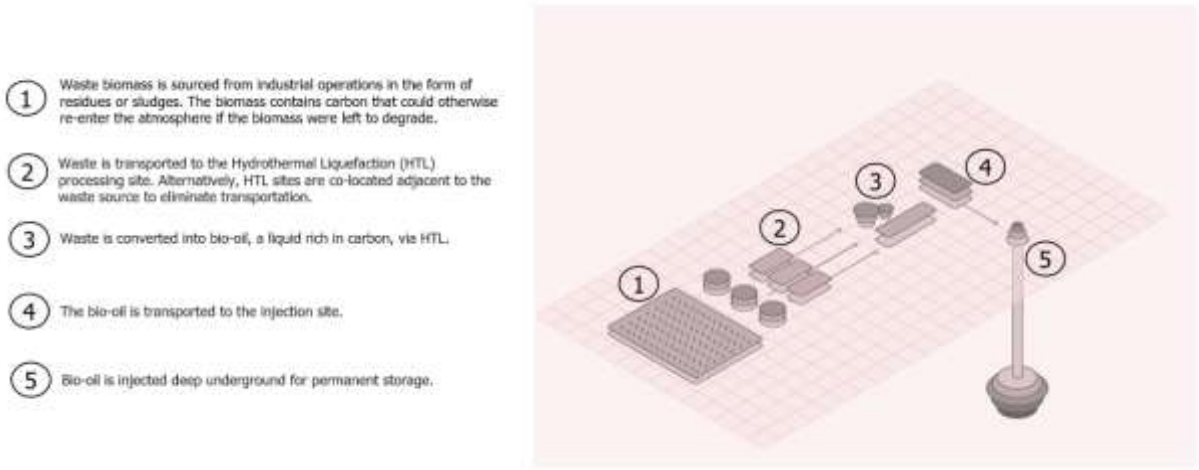
in PDD	
Details and links to any supporting documentation	Uploaded to certify or to the Validation and Verification Body via the shared Google drive folder.
Please confirm that the GHG Statement has been undertaken in accordance with the information provided in the project boundary section of the PDD. Provide any additional information relating to the GHG Statement project boundary here.	Yes

Section B – Project Details

B. GHG Statement - Project Details
Provide information on the Project design and implementation concerning the reporting period. *This should include the following information as a minimum:* ● *Key features of the Project including a technical description* ● *information on the implementation status of the project, For the description of the installed technologies, technical processes and equipment, include diagrams, where appropriate*
NULIFE's Hydrothermal Liquefaction (HTL) process is a continuous thermo-chemical process that breaks down and liquefies high-moisture organic materials. The exothermic reaction occurs under high pressure (2,900 – 3,700 psig) and temperature (290 - 375°C), using water as a catalyst to reform the biomass carbon into long-chain hydrocarbons (bio-oil). Carbon Removal Steps: 1. Dry biomass waste is delivered and stored on site. Wet biomass is delivered and stored in 10,000L storage tanks. 2. The wet biomass feedstock is pumped and stored into holding tanks in preparation for the HTL process. The slurry can be mixed with additional dry biomass and/or water to obtain the desired solid content or can be processed "as received". 3. The slurry mixture moves through a series of pumps into the processor. The primary pump generates the necessary pressure for the HTL reaction (2,900 – 3,700 psig). 4. Once pressurized, the slurry travels through the HTL processor. The current processor heat source is electric heaters. The process operates at 290 °C to 375 °C. 5. After the reaction takes place, the effluent material discharges through a pressure letdown valve. 6. The effluent products are released into the flash tank. The bio-oil (with entrained biochar solids) and effluent water are transferred into the external bio-oil storage tank. The bio-oil accumulates until there is sufficient volume for a shipment to the injection site. 7. The wastewater collected from the separation of the bio-oil is transferred into the external water storage tank. In the Saskatoon plant, this wastewater can be recycled into the front of the process or sent for safe treatment/disposal offsite.

8. From the flash tank, the gas phase passes through an activated carbon canister and is then vented into the atmosphere. The resulting gas phase has been validated to be 99% biogenic CO_2, which will be released back into the atmosphere via the vent stack. Currently, this CO_2 is not captured. In the future it could be collected and repurposed for industrial applications to offset fossil CO_2 or injected underground for carbon dioxide removal.
9. The bio-oil liquid is transported to the injection site by a third-party road transport company.
10. Upon arrival, the bio-oil is offloaded by the transport company at the injection site offloading pad. The truck offloads the bio-oil into the injectant tank through a connected hose.
11. The site operator then pumps the bio-oil underground into the salt cavern.
12. As bio-oil is injected, it displaces existing brine within the cavern. Brine is removed and is stored in brine tanks. If solid material is identified in the brine tank, it is removed and re-injected into the salt cavern.
13. Clean brine is discharged into a permitted brine disposal well onsite.

Project boundary Diagram:



Please refer to the PDD for the project details. Currently there are no additional details beyond what is included in the PDD.

Section C – Project Changes

C.1 Summary of changes

Please provide details of any changes or deviations from the planned implementation, operation, or monitoring of the project activity that occurred during this reporting period. Project Proponents must determine whether the changes are temporary deviations or permanent changes.
Temporary deviations could include deviations from:

a) *The monitoring plan mentioned in the approved PDD.*
b) *The applied Protocol.*
c) *The applied modules.*
d) *Data collection procedure.*
e) *GHG Statement Methodology.*
f) *Uncertainty assessment.*

If there were any temporary changes or deviations corresponding to this reporting period, please include the details of those changes in this section. If the changes do not fall into the above categories, please note them in Section F of the corresponding PDD instead.

(If there were no deviations during the reporting period, this section is not applicable. In that case, kindly leave this section blank and mention "Section is intentionally left blank.")

No changes.

C.2 Impact of changes

For each deviation, provide a description of the nature, extent and duration of the non-conforming reporting period, and:

a) *Propose alternative monitoring arrangements for the non-conforming reporting period; or*
b) *Apply the most conservative values approach*

(If there were no deviations during the reporting period, this section is not applicable. In that case, kindly leave this section blank and mention "Section is intentionally left blank.")

None of the changes listed above have impact on this GHG Statement report quantification. They are presented to introduce the upcoming changes to the verifier.

Table: 1 Temporary changes to the project

Period of change	*Nature of change*	*Impact of change*

Section D – GHG Statement Methodology

D.1 GHG Statement Methodology

Provide information on the GHG Statement approach and methodology in relation to data collection.

This should include the following information as a minimum:

- *Data collection procedures - how data was requested and gathered, who supplied each data point, the format data was received in, and the date data points were received*
- *How data quality was assessed and how the data hierarchy was followed in terms of using measured, calculated and estimated data. Evidence and justification should be provided in any cases where measured data was not used*
- *Details of data validation processes followed, including treatment of missing data*

Evidence of all raw data that informed the assessment (including data that informed any assumptions) must be appended to the PDD, or linked within the GHG Statement.

This GHG statement contained 4 removals:

rmv_1M0AYD0T71S0YPEZ
rmv_1M0B2H1GT1S0YSRW
rmv_1M0B3MSMK1S0Y71X
rmv_1M0B59RM61S0BW9G

Biomass Feedstock - Counterfactuals
The biomass feedstock used for this reporting period is an effluent sludge from two regional canola crushing plants. More details of the feedstock can be found in the Feedstock section of Certify.

Data Collection
Details on the specific Emissions Factors (EF) and scientific values used can be found as attachments in Isometric Certify software in the LCA section along with the PDD Section B.4 Project Boundary.

Details on specific calculation formulas can be found in Isometric Certify software in the LCA section.

<u>Project Emissions (HTL)</u>

- As per Section 3.3 of the Embodied Emissions Accounting Module, the mass-based method has been selected: Overall mass of equipment, separated by material type, and material emissions factors.
- As per Section 3.2 of the Embodied Emissions Accounting Module, emissions are allocated over the project lifetime, which is 25 years.
- For the HTL conversion site, the mass of steel, stainless steel and polyethylene plastic are determined based on equipment lists.
- Embodied emissions are calculated based on the mass x EF formula.
- Supplier locations are considered to calculate transport emissions using the mass x distance x EF formula.
- End of life scenarios are applied for each material using the mass x EF formula.

Sequestrations - Bio-oil storage

Bio-oil - Wt % of Carbon

- This value is primary measured data.
- Carbon wt % of bio-oil results are delivered via raw equipment data using Microsoft Excel. Data is uploaded to OneDrive through an external sharing folder between NULIFE and the lab. The data is then copied into a master excel spreadsheet on OneDrive where the weighted average carbon wt% of the samples is determined.
- Carbon wt% Input is utilized for GHG Statements in Certify to determine “bio-oil storage” for the relevant “removal data”.
- The relevant Input is labelled “carbon content of product.”
- Certify Uploaded Source: Excel file of the Carbon wt% data including the calculated average that is used for this input.

Bio-oil Carbon Content Measurement: Method A – measure every batch (Sampling the blended injection batch)

Example of how “carbon content of product” is calculated:
Sample 1: average carbon wt% of 75%
Sample 2: average carbon wt% of 73%
Sample 3: average carbon wt% of 74%

Math: (0.75 + 0.73 + 0.74)/3 = 0.74
0.74 value is used for the carbon content of the product in this example.

Bio-oil - Mass of product

- This value is primary measured data.
- Liters of bio-oil production are measured and recorded as a cubic meter value.
- Moisture content of bio-oil data is calculated.
- Feedstock specific Bio-oil density data is calculated.
- Using the density of the bio-oil, the cubic meter value (moisture deducted) is converted into kilograms.
- Mass Input is utilized for GHG Statements in Certify determine “bio-oil storage” for the relevant “removal data”.
- Mass Input is utilized for GHG Statements in Certify determine “bio-oil storage” for the relevant “removal data”.
- The relevant Input is labelled “mass of product”.
- Certify Uploaded Source: Excel file of the total bio-oil mass for the relevant injection batch.

Removal Activities - Bio-oil Production

Electricity usage power meter

- This value is primary measured data.
- Power Meter logs are recorded on the Programmable Logic Controller (PLC). The unit of measure is kWh.
- Electricity usage is recorded as a kWh value over the course of the relevant operational dates for the removal. The power meter log results in a single kwh value for the usage over a specific removal period.
- The kWh value is entered for the input “Grid electricity usage”.
- Since Renewable Electricity Certificates (REC) are acquired on a volumetric matching basis, the kWh value is entered a 2nd time for the input “procured power electricity usage”.
- Certify Uploaded Source: PLC software evidence of kWh consumption.
- Certify Uploaded Source: REC procurement evidence.

Tailgas methane emissions - total mass flow of gas

- This value is primary measured data.
- The mass flow rate of the tailgas is calculated on a per HTL production batch basis by deducting the output quantities of bio-oil, water soluble oil and biochar.
- The mass input is utilized for GHG statements in Certify to determine “tailgas methane emissions” for the relevant “removal data”.
- The input “total mass flow of gas” is entered as a “kg” unit of measure.
- Certify Uploaded Source: Excel file of the total bio-oil mass for the relevant injection batch.
- The appropriate scientific value for Methane (CH4) is utilized for the mass flow x scientific value formula.

Wastewater Transport emissions (tanker) - Distance Travelled
Distance of effluent wastewater transported from conversion site to disposal site

- This value is primary measured data.
- Distance is determined via Google Maps.
- Distance input is utilized for GHG statements in Certify to determine “wastewater transport emissions” for the relevant “removal data”.
- The unit of measure is km.
- Certify Uploaded Source: Mapping screenshot i.e. Google Maps or transport documentation.

Wastewater Transport emissions (tanker) - Mass of Load
Mass of effluent wastewater transported from conversion site to disposal site.

- The transport company provides a bill of lading invoice with tanker total mass.
- Mass data is transferred into Excel File: Incoming Outgoing Delivery Log Ontario Avenue .
- Multiple event mass values are combined into a single value for the removal.
- Mass input is utilized for GHG statements in Certify to determine “wastewater transport emissions” for the relevant “removal data”.
- The unit of measure is kg.
- Certify Uploaded Source: Transport Bill of Lading, invoice or other documentation

The GLEC mass x distance tanker emissions factor includes the return trip impacts.

<u>Removal Activities - Bio-oil Storage</u>
Equipment Embodied Emissions per MT of injectant

- As per Section 3.2 of the Embodied Emissions Accounting Module, storage infrastructure is shared amongst multiple entities thus Project Proponent must allocate a percentage of the embodied emissions of that storage infrastructure proportional to the mass of stored material to the total storage capacity of that infrastructure.
- For the injection site, the mass of equipment is determined based on equipment lists.
- This value is primary measured data.
- Injection site equipment embodied emissions. - An emissions factor to determine Mass based CI emissions is attached in Certify.
- Based on the equipment embodied emissions and a total facility storage capacity, a kg CO2e/kg of Injectant factor is calculated.
- The injectant Mass is determined based on the invoice from the injection site and appropriate documentation such as the weight scale ticket.
- Mass input is utilized for GHG statements in Certify to determine “equipment embodied emissions per MT of injectant” for the relevant “removal data”.
- The unit of measure is kg.

- Certify Uploaded Source: for the kg CO2e/kg of Injectant factor is an excel spreadsheet.
- Certify Uploaded Source: Injectant mass is determined using the injection site weight scale ticket.

Injection electricity usage
- This data is calculated.
- Injection site electricity usage.
- The unit of measure is kWh.
- The Injection Partner has provided specifications on their primary pump and its horsepower along with frequency the pump is running (hours) on a per cubic meter of injectant basis.
- kWh input is utilized for GHG statements in Certify to determine “injection electricity usage” for the relevant “removal data”.
- Certify uploaded source: excel spreadsheet calculating the kWh per injection value based on the pump specifications and the hours of pumping required for the injection event.

<u>Removal Activities - Bio-oil Transport</u>
Bio-oil transport to storage site via truck (tanker) - Distance travelled
- This value is primary measured data.
- Distance of bio-oil transported from conversion site to injection site.
- Distance is determined by using Google Maps.
- The data is uploaded directly to Certify.
- Distance input is utilized for GHG statements in Certify to determine “bio-oil transport to storage site via truck” for the relevant “removal data”.
- The unit of measure is km.
- Certify uploaded source: Mapping software evidence or bill of lading.

Bio-oil transport to storage site via truck (tanker) - Mass of load
Mass of bio-oil transported from conversion site to injection site
- This value is primary measured data.
- The injection site provides an injection statement including ticket with delivery mass based on arrival and departure weight.
- The data is uploaded to Certify.
- Mass input is utilized for GHG statements in Certify to determine “bio-oil transport to storage site via truck” for the relevant “removal data”.
- The unit of measure is metric tonne.
- Certify uploaded source: weight scale ticket from injection site.

The GLEC mass x distance tanker emissions factor includes the return trip impacts.

<u>Removal Activities - Biomass sourcing</u>
Biomass feedstock delivery event emissions are including with 100% allocation during the reporting period they fall in.

Biomass (dry/wet) transportation to production site via truck (flat deck/tanker) - Distance travelled
- This value is primary measured data.
- Distance of biomass transported from supplier to conversion site
- The location of the supplier is recorded on bill of lading or other documentation.
- Distance is determined by using Google Maps.

- The data is uploaded directly to Certify.
- Distance input is utilized for GHG statements in Certify to determine “biomass transport to storage site via truck” for the relevant “removal data”.
- The unit of measure is km.
- Certify uploaded source: Mapping software evidence or bill of lading.
- The allocation approach may be used to separate biomass transport emissions between removal batches.

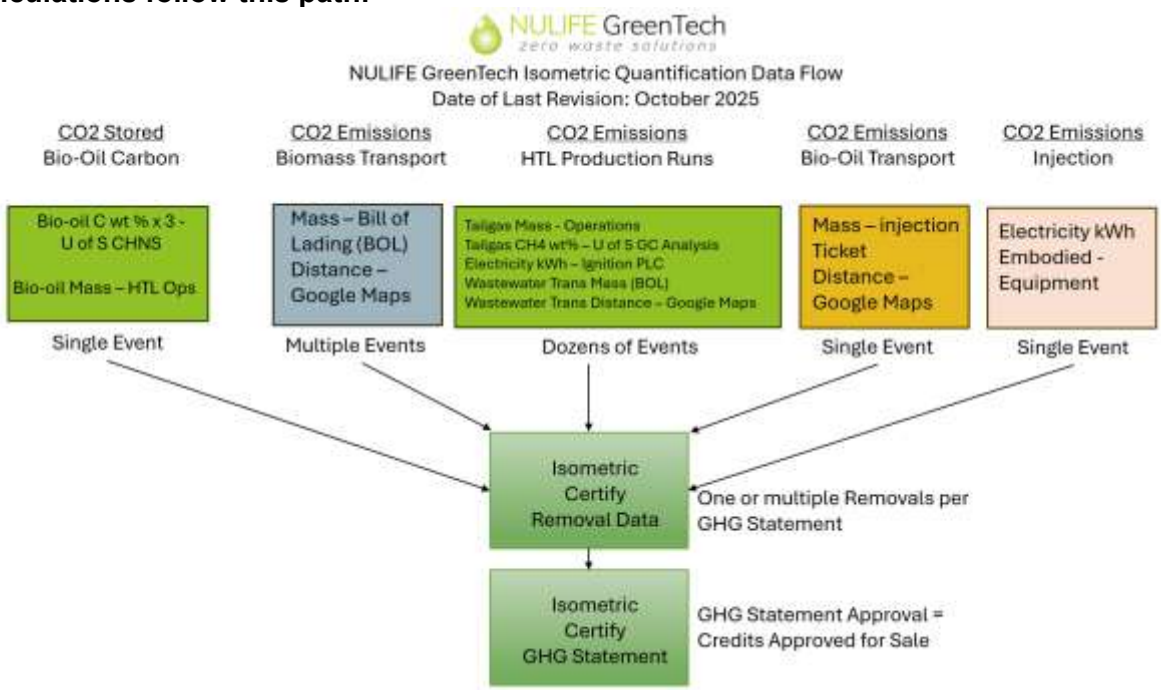
Biomass (dry/wet) transportation to production site via truck (flat deck/tanker) - Mass of load

- This value is primary measured data.
- Mass of biomass transported from supplier to conversion site
- The transport company provides a bill of lading with delivery mass provided by the feedstock supplier.
- The data is transferred to Certify.
- Mass input is utilized for GHG statements in Certify to determine “biomass transport to storage site via truck” for the relevant “removal data”.
- The unit of measure is kg.
- Certify uploaded source: Transport event documentation i.e. bill of lading or invoice.
- The allocation approach may be used to separate biomass transport emissions between removal batches.

The GLEC mass x distance tanker/flat deck emissions factor include the return trip impacts.

Please refer to section B9 Monitoring Plan and B10 Data Collection and Storage for the more relevant information to this section.

Calculations follow this path:



CO2 Stored

- Bio-oil Mass x Bio-oil weighted average carbon wt% value x Scientific Value.
- The scientific value of 3.67 CO_2e eq mass of C is used to convert bio-oil carbon into

CO_2.

CO2 Emissions
Biomass/Bio-oil/Wastewater Transport: Mass x Distance x EF

Tailgas
- Tailgas mass x methane (CH4) Scientific Value.

Electricity at HTL site
- kWh consumed x grid electricity EF
- kWh consumed x clean power procurement EF

Electricity at Injection site
- kWh consumed x grid electricity EF (no clean power procurement)
- Embodied emissions: Mass x EF

Sensitivity analysis is automated in Isometric's Certify software and details will be provided there.

Section E – GHG Statement Results

GHG Statement Results

The following information should be provided in this section:
- *Baseline results reported in t CO_2e for the Reporting Period*
- *Net CO_2e removals results reported in t CO_2e for the Reporting Period. These must be aggregated and also broken down into GHG SSRs*
- *If it has not already been covered in the '*Uncertainty assessment' section of the PDD, provide *outcomes of sensitivity and uncertainty analyses including a statement of how uncertainty affects the results and how it has been addressed to minimize misrepresentation*
- *A statement of how the assumptions and choices made in the assessment are conservative*

The GHG Statement calculations must be provided separately, including all raw data and evidence. The GHG Statement calculations must be clearly referenced with a transparent audit trail of evidence, decision making, assumptions, explanations.

Baseline Results
Total Tonnes removed 150.45 tCO_2e. See Isometric Certify for a breakdown of the results. The calculation view of each reomval contains the breakdown of carbon sequestered and the emissions.

Uncertainty Assessment
The Sensitivity Analysis flagged four items with an impact on net removal of > 20%. These items are mass of product (bio-oil) and Carbon content of product (bio-oil). Both values provide uncertainty justification:

Mass of Product: The datapoint is accurate primary data and does not require an uncertainty discount. Bio-oil mass is measured data, with high certainty. Original values are used without a standard deviation.

Carbon content of product: The datapoint is accurate primary data and does not require an uncertainty discount. Carbon wt% is measured data, with high certainty due to the elemental analysis equipment operational requirements of utilizing standards to confirm accuracy of equipment, the frequency of sampling and duplicate samples being tested. Each HTL Injection batch is tested for carbon wt% with three (3) unique samples of bio-oil. Each sample is tested in duplicates in the elemental analysis machine to determine the carbon wt% value.

Grid Electricity/Procured Power Electricity Usage: The datapoint is accurate primary data and does not require an uncertainty discount.

The remaining data points do not impact net removals by greater than 1%, thus further analysis is not required.

Conservative Assumptions
Assumptions and choices made in the assessment are conservative.

Nearly all the data points are primary measured data including:

- Bio-oil Mass,
- Bio-oil Carbon wt%,
- Wastewater Mass,
- Tailgas Mass,
- HTL Electricity Usage,
- Biomass Mass.

At the injection site, a 20% increase in electricity usage is assumed due to the lack of primary measured data from the injection site.

Transport event distances are based on factual routes between points A and B. The mass x distance emissions factors include the return trip impacts. Additionally, a 5% increase in transport distance is added via standard deviation to account for potential extension of routes.

Materiality Assessment
Certain SSRs were omitted from this removal due to the Materiality Assessment exercise, as per GHG Accounting V1.0. More details are below:

Title: Materiality Assessment				
Date: September 10, 2025				
Protocol: GHG Accounting V1.0				
Data Used: Isometric's Certify Net Negativity NULIFE Example				
CO2e Removal (tCO2e)	4188			
If materiality of the SSR is < 1%, then the emissions associated with the SSR may be excluded from the GHG Statement boundary, provided that the total exclusions make up < 1% collectively. If the sum of SSRs that individually are < 1% becomes > 1%, then the SSRs with the highest Materiality must be included in the GHG Statement boundary until the sum of SSRs becomes < 1%. It is not necessary to calculate emissions related to SSRs that are < 1% using more accurate data if the emission calculations for the Materiality assessment were conducted using a high-level approach.				
The estimated total net CO2e Removal should be calculated using actual data and following all required methodologies for SSRs considered to be material.				
SSR	SSR tCO2e	% of CO2e Removal	Included/Excluded?	Notes
Biomass (wet) transportation to production site via truck (tanker)	0.005	0.119388073	Included	Important SSR that we will continue to include
Biomass (dry) transportation to production site via truck (flat deck)		0	Included in future	example did not include dry biomass. Will include SSR when events take place
Wastewater treatment emissions	0.003	0.071633238	exclude	
Clean Power procurement (REC) Power Meter	0.092	2.196752627	Included	HTL Site - Will include SSR when events take place
Tailgas methane emissions	0.03	0.716332378	Included	HTL Site - Will include SSR when events take place
Municipal water consumable	0.001	0.023877746	exclude	
Clean power procurement (REC) equipment manual log	0.005	0.119388073	exclude	
Clean power procurement (REC) usage analytical equipment	0.001	0.023877746	exclude	
Hydrogen Peroxide Consumable	0.009	0.214899713	exclude	
Sodium Hydroxide Consumable	0.001	0.023877746	exclude	
Activated Carbon Consumable	0.003	0.071633238	exclude	
Wastewater Transport emissions (tanker)	0.002	0.047755492	Include	Will include SSR when events take place due to potential for change in transport distance
Activated Carbon Transport emissions (flat deck)	0.001	0.023877746	exclude	
Hydrogen peroxide/sodium hydroxide transport emissions (flat deck)	0.001	0.023877746	exclude	
Bio-oil transportation to storage site via truck (tanker)	0.038	0.907354346	Include	Will include SSR when events take place
Equipment Embodied Emissions per MT of injectant	0.001	0.023877746	Include	Will include SSR when events take place
Injection electricity usage	0.007	0.167144222	Include	Will include SSR when events take place
Staff travel sample delivery (gasoline car)	0.001	0.023877746	exclude	
staff travel operators to site	0.007	0.167144222	exclude	
Sum of SSRs excluded (must be < 1%)	0.208	**0.787965616**		still below the 1% threshold

Section F – GHG Statement Version Changes

GHG Statement Version Changes

For each statement resubmission, please specify the new version date (which is the same as the resubmission date) and the updated total of CO_2e tonnes removed.

GHG Statement Versions

Version date	*Total CO_2e tonnes removed*
August 19th, 2026	150.45